Heart Disease Prediction Using Machine Learning (Random Forest Approach)

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SUBJECT- AIML PROJECT

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1. Problem Statement

Heart disease is one of the leading causes of death worldwide. Early detection is critical but often requires multiple medical tests, making it time-consuming and expensive.

This project aims to solve the problem of early risk prediction by building a machine learning model that can classify whether a person is likely to have heart disease based on medical attributes.

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2. Why This Problem Matters

- Heart disease often develops silently
- Early prediction can reduce mortality
- Healthcare systems benefit from decision-support tools
- Useful for preventive healthcare and awareness

This problem is relevant as it combines healthcare impact with data-driven decision making.

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3. Approach

The solution follows a structured machine learning pipeline:

Step 1: Data Collection

- Dataset sourced from Kaggle
- File used: heart_disease_uci.csv

```
• path = kagglehub.dataset_download("redwankarimsony/heart-disease-data")
```

Step 2: Data Cleaning

- Removed irrelevant columns (id, dataset)
- Renamed num -target
- Replaced missing values (?) with null

Step 3: Data Preprocessing

Different strategies were used based on feature types:

- **Numerical Features**
  - Converted to numeric
  - Missing values filled using median
- **Binary Categorical Features**
  - Converted to 0/1
  - Missing values filled using mode
- **Multi-Categorical Features**
  -Missing values filled using mode
  -One-hot encoding applied

Step 4: Feature Engineering

- Target converted to binary classification:
  - 0 -No disease
  - 1 -Disease

Step 5: Model Training

- Algorithm: Random Forest Classifier
- Data split:
  - 80% training
  - 20% testing
- Stratified sampling used to maintain class balance

Step 6: Feature Scaling

- StandardScaler has been applied to normalize features

Step 7: Evaluation

Metrics used

  - Accuracy
  - Precision
  - Recall
  - F1-score

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4. Key Decisions Made

- Used median imputation instead of dropping rows as it preserves data
- Used mode for categorical values for most logical replacement
- Applied one-hot encoding for better model interpretation
- Used Random Forest as it is robust and beginner-friendly
- Used stratified split to avoid class imbalance bias

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5. Challenges Faced

1. Missing Values

- Dataset contained inconsistent missing data (?) which required multiple preprocessing strategies

2. Mixed Data Types

- Some columns had numerical and categorical formats which needed careful conversion

3. Feature Encoding

- Deciding which features to encode and how

4. Small Dataset

- Limited data increases risk of overfitting

6. Results

- Accuracy achieved: 83%
- Model performs reasonably well on unseen data
- Balanced performance across classes

```
--- Model Evaluation ---
Accuracy: 0.8208

Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.76      0.79        78
           1       0.81      0.87      0.84        95

    accuracy                           0.82       173
   macro avg       0.82      0.82      0.82       173
weighted avg       0.82      0.82      0.82       173
```

7. Limitations

- Small dataset size (300 rows)
- No hyperparameter tuning
- No cross-validation
- Model not validated on real-world data

8. What I Learned

- Importance of proper data preprocessing
- Handling missing values intelligently
- Difference between categorical and numerical features
- How ensemble models like Random Forest work
- Why evaluation metrics matter beyond accuracy
- Importance of avoiding data leakage

9. Future Improvements

- Apply cross-validation
- Perform hyperparameter tuning

- Use advanced models (XGBoost, LightGBM)
- Add explainability (feature importance)
- Deploy as a web application

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10. Conclusion

This project successfully demonstrates the use of machine learning in predicting heart disease. While the model shows good performance, it is intended as a support tool and not a replacement for medical diagnosis. With further improvements and validation, such systems can contribute to preventive healthcare solutions.